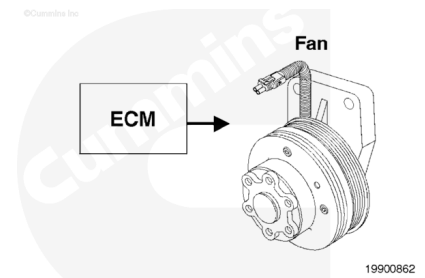


Trainings ▾

General Information

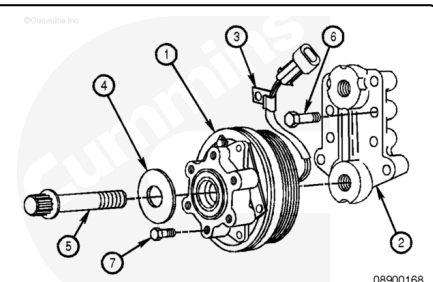
⚠ WARNING ⚠

The cooling fan will engage when the engine is started. To prevent serious personal injury, keep your hands clear of the fan's path when starting the engine and during engine operation.



The Cummins®-developed electromagnetic fan clutch product is an integrated package with the clutch, bracket, shaft, bearings, pulley, and fan spacer designed as a unit. An engine coolant sensor and/or a refrigerant pressure sensor (if equipped with air conditioning) will engage and disengage the clutch. The fan clutch engages when the coolant temperature sensor or the refrigerant switch is closed. It disengages when either of these sensors returns to the open position. The most common temperature switch is set to close at 91°C [195°F]. Contact the OEM for recommended temperature switches. The temperature **must** be set above the opening point of the thermostat. When cooling is needed, a powerful electromagnet engages the fan. The fan is either fully engaged or fully disengaged. The OEM could have a manual override switch to engage the clutch manually at the operator's command. The electromagnetic fan clutch is driven by a poly-vee drive belt and is available with an accessory drive for vee-drive belts.

Clutch wires **must** be oriented at 2 o'clock instead of 12 o'clock. The following table lists the component parts of the electric fan clutch.



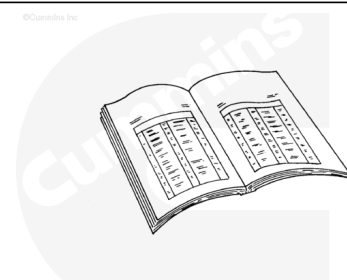
The following parts have been released for B Series production and service:

Parts Breakdown				
	Part Number	Part Name	Description	Comments
1	3922954	Clutch	B Series with Accessory Drive	1.25:1 Ratio
1	3922773	Clutch	B Series without Accessory Drive	1.25:1 Ratio
2	3922869	Bracket	Fan Center 343 or 393.5 mm [13.5 or 15.5 in]	1 per Assembly
3	3920348	P-Clip		1 per Assembly
4	3922952	Washer		1 per Assembly
5	3922953	Capscrew 12 Point		1 per Assembly
N/A	3862596	Warning Label *		Per Installation
6	3900678	Capscrew		4 per Assembly
7	3903464	Fan Bolts	40-mm [1.6-in] length	6 per Assembly
7	3900634	Fan Bolts	50-mm [1.9-in] length	6 per Assembly
7	3904446	Fan Bolts	70-mm [2.8-in] length	6 per Assembly
7	3900067	Fan Bolts	80-mm [3.15-in] length	6 per Assembly
7	3916069	Fan Bolts	90-mm [3.54-in] length	6 per Assembly
* Each 3862496 warning label kit contains two labels. One label must be installed on both sides of the fan shroud.				

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

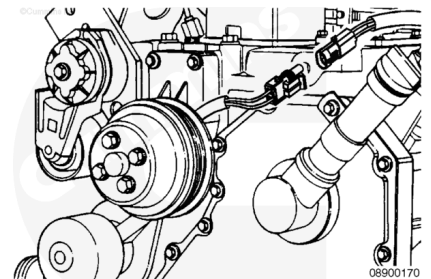


ck800wa

- Disconnect the batteries. Refer to the OEM service manual.
- Remove the cooling fan and spacers. Refer to the OEM service manual.
- Remove the drive belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html>)

Remove

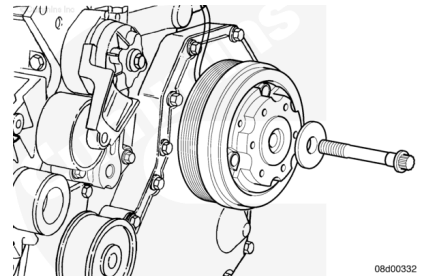
Disconnect the fan clutch connector on the base harness from the fan clutch.



Note : As the electric fan clutch mounting fastener is loosened, make sure to support the clutch.

Note : The electric fan clutch mounting capscrew has an external Torx™ head.

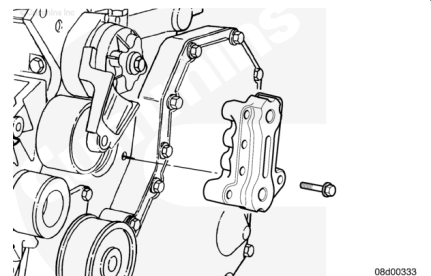
Remove the electric fan clutch mounting capscrew, washer and electric fan clutch.



Note : Prior to removing the electric fan clutch support bracket, note the mounting location and orientation of the bracket. The mounting bracket can be installed in different orientations for different fan drive arrangements.

Note : Note the location of the P-clip for the electric fan clutch wiring harness pigtail.

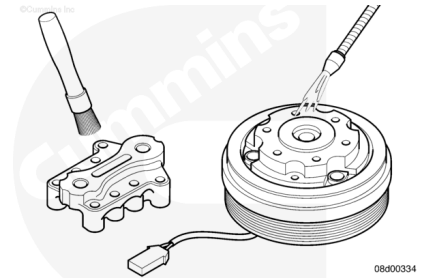
Remove the electric fan clutch support bracket and mounting capscrews.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



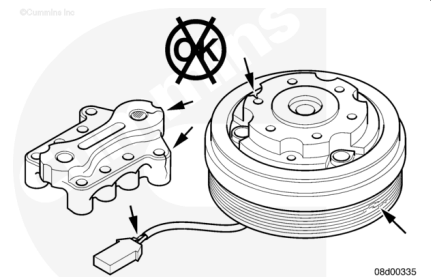
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the fan clutch and bracket with solvent.

Dry with compressed air.

Inspect the fan clutch and bracket for damage.

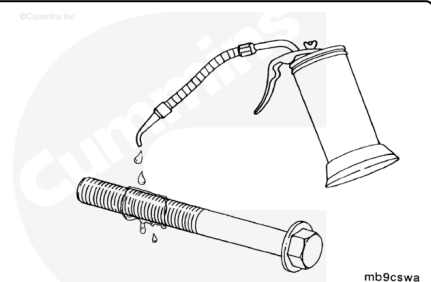


Install

Do **not** use Loctite™ or any other thread locking compounds. Make sure capscrew head surface and threads are clean and free of debris.

Use new capscrew if any damage is found.

Lubricate the threads and underside of the capscrew head with clean 15W-40 engine lubricating oil.

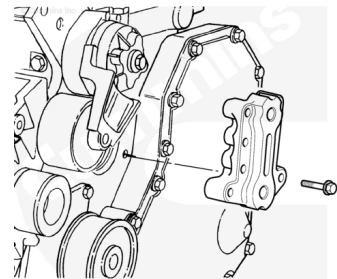


Note : Make sure to install the electric fan clutch mounting bracket in the same orientation and location as removed. The mounting bracket can be installed in

different orientations for different fan drive arrangements.

Install the electric fan clutch support bracket, wiring harness pigtail P-clip and mounting capscrews.

Torque Value: 33 n•m [24 ft-lb]



Note : The electric fan clutch mounting capscrew has an external Torx™ head.

Install the electric fan clutch, washer and mounting capscrew.

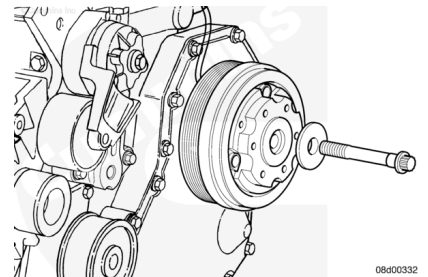
Prior to tightening the capscrew, rotate the fan clutch so that the wires coming out of the back of the fan clutch are captured by the P-clip. Bend the P-clip over by hand to secure the wires.

Note : Make sure that wires are not being pulled at the fan clutch.

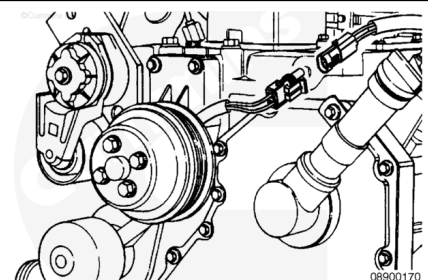
Tighten the fan clutch mounting capscrew.

Torque Value:

1. 102 n•m [75 ft-lb]
2. Rotate 60 degrees



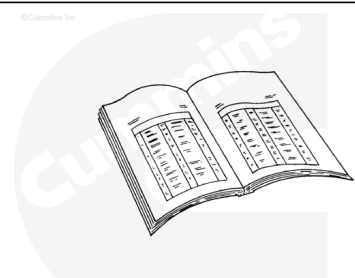
Connect the fan clutch connector on the base harness to the fan clutch.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last



- Install the drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html)
- Install the cooling fan and spacers. Refer to the OEM service manual.
- Connect the batteries. Refer to the OEM service manual.

Last Modified: 18-Apr-2012
